

R&D AI Uploader

Complete Scenarios & Edge Cases

All 128 scenarios, rules, and guardrails — from file intake to field-level protection

128

Total Scenarios

12

Prompt Sections

13

Upload Scenarios

30

Guardrail Rules

Occams Advisory

R&D Tax Credit Qualification Tool | AI Auto-Fill Agent | Version 3.0

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PART 1 — FILE INTAKE RULES (5 Rules)

These rules govern what the agent must do before extracting a single field. Executing these first prevents cascading errors downstream.

1 F-1 — Read Every File Before Creating Any Record

Process ALL uploaded files first. Build a complete inventory. Only then begin creating records. File 1 may have the project name. File 3 may have the hours. File 7 may have the employee list. Creating a record after reading only File 1 leads to incomplete records and false duplicates.

- Read all files → build full inventory → then create records
- Never commit to a project record mid-batch

2 F-2 — Identify the File Type and Its Role

Each file plays a different role: Job Cost Summary PDFs provide phase-level hour totals; Job Cost Detail PDFs provide line-by-line transactions; Payroll Reports provide employee-level breakdown; All-Cost Detail Reports show every cost type; Contracts and Word docs provide narrative content.

- PDF cost report → hours, employees, supplies
- Word proposal/SOW → description, challenges, solutions
- Contract PDF → contract type, funded status, client name
- Unrecognized file → extract what is parseable; do not guess

3 F-3 — Extract Deduplication Keys from Every File

Before doing anything else, extract from every file: (1) Project Identifier — Job#, Project#, Contract#, or Product Code; (2) Project Name — human-readable name; (3) Company Name; (4) Date Range; (5) Report Type. Record these on an inventory card for every file before proceeding.

- 5 keys per file: Identifier, Name, Company, Date Range, Report Type
- Create an inventory card per file — never skip this step

4 F-4 — Handle Files That Cannot Be Read

If a file fails to open, is password-protected, is a scanned image with no text layer, or produces garbled output: do NOT create a record, do NOT hallucinate data, flag it explicitly, continue processing all other files. If it is the ONLY file uploaded → return zero records with explanation.

- Unreadable file → flag + skip, never guess content
- Single unreadable file → 0 records, request re-upload

5 F-5 — The 10-File Upload Limit

The system accepts up to 10 files per batch. Process the first 10. Notify the user if more are present: 'Only the first 10 files were processed. Please upload remaining files in a separate batch.'

- Max 10 files per batch
- Excess files → notify, do not silently ignore

PART 2 — PROJECT IDENTITY RULES (7 Rules)

The most critical section. These rules define what counts as a 'project' and prevent phases, recap rows, and sub-tasks from being mistaken for projects.

6 P-1 — THE MASTER RULE: One Project Identifier = One Record

A project is defined by its unique identifier (Job Number / Project Number / Contract Number). Not by its name. Not by its files. Not by its phases. Same identifier across 10 files → 1 record. Different identifiers in 1 file → multiple records.

- Identifier is the deduplication key — not name, not file count
- This rule overrides everything else. Apply it first, always.

7 P-2 — The Three-Level Document Hierarchy

Construction and engineering cost reports use: Level 1 = PROJECT/JOB (creates one record); Level 2 = PHASE/WORK PACKAGE (sub-task, never creates a record); Level 3 = TRANSACTION/EMPLOYEE ENTRY (line item, never creates a record).

- Job 314 — Coyote Flood Management = Level 1 → 1 record
- Phase 032030: Seal Paired Sheetpile Joint = Level 2 → NO record
- PR GILJES / Jose Jesus Gil / 8.00 hrs = Level 3 → NO record

8 P-3 — How to Identify Level 1 (the Project) in a Document

Look in the report header block for: Job: [number] Phase: ALL Employee: ALL. Then in the data section for: Job: 314 COYOTE FLOOD MANAGEMENT. Then in the footer for: Total for Job: 314 6,059.50 hrs \$572,728.26.

- Report header block: Job: 314 → identifier
- First data row: Job: 314 COYOTE FLOOD MANAGEMENT → name
- Footer recap: Total for Job: 314 → verification

9 P-4 — How to Identify Level 2 (a Phase) — Never Create Records From These

Phases appear as: Phase: 032030 Seal Paired Sheetpile Joi Cost Type: L LABOR. The test: Does the line have a 4–6 digit numeric Phase Code before the description? Yes → Phase → NO record. Real examples: 032030 Seal Paired Sheetpile Joint, 601100 Project Superintendent, 885100 FENCE ADDITION, 915100 RESTORATION IMPROVEMENTS — ALL phases, never projects.

- 4–6 digit code before description = Phase = no record
- Follows 'Phase:' label = Phase = no record

10 P-5 — Project Title Normalization

Once you have the Job Number and Name, normalize to: 'Job [NUMBER] - [Title Case Name]'. Examples: 'Job: 314 COYOTE FLOOD MANAGEMENT' → 'Job 314 - Coyote Flood Management'. Expand abbreviations (Mgmt → Management, Rehab → Rehabilitation). Never include phase names, cost type labels, or change order suffixes in the title.

- Lead with identifier, dash, then name in Title Case
- Never append phase codes, cost types, or change order names

11 P-6 — Change Orders and Scope Additions Are NOT Separate Projects

A change order expands an existing project. Same Job Number + suffix = fold into base record. Signals: same Job# as base, suffix like 'FENCE ADDITION / ARCHITECTURAL ADDS / RESTORATION IMPROVEMENTS', document says 'Change Order #3 to Contract 314', or name is [BASE NAME] - [SCOPE ADDITION].

- 'Job 314 - COYOTE FLOOD MANAGEMENT - FENCE ADDITION' → fold into Job 314
- Add change order hours to base record; note scope expansion in description

12 P-7 — Name Variations Are the Same Project

The same project may appear with different names across files. The Job Number is authoritative. COYOTE FLOOD MANAGEMENT, COYOTE FLOOD MGMT, 314 COYOTE FLOOD MANAGEMENT, Job 314 - Coyote Flood Management - Native Riparian Habitat Restoration — all the same project.

- Job Number match → same project regardless of name variation
- Normalize all variations to the canonical title

PART 3 — MULTI-FILE UPLOAD SCENARIOS (13 Scenarios)

Every combination of files a user can upload. The agent must handle each scenario correctly and produce exactly the right number of records — never more.

13 Scenario A — Single File, Single Project

User uploads 1 file. It contains data for 1 project. The file may be 80+ pages — still 1 project. Recap/summary pages at the end repeat totals — do NOT create a second record for the recap.

Expected Output: 1 record

14 Scenario B — Multiple Files, Same Single Project (Most Common)

User uploads 2–10 files all referencing the same Job/Project Number. Different report types (payroll, cost detail, summary) about the same project. Real example: 4 files all for Job 314. Merge: hours from labor report, employees deduplicated across all files, description synthesized.

Expected Output: 1 record

15 Scenario C — Multiple Files, Multiple Different Projects

User uploads files for completely different projects in one batch. Example: Files 1–3 = Job 314, Files 4–5 = Job 287, Files 6–7 = Job 401. Never merge data from different Job Numbers. Each record is completely independent.

Expected Output: 1 per Job#

16 Scenario D — Multiple Files, Mix of Project Data + Supporting Documents

User uploads cost reports for one project (3 files) + a Word technical scope (1 file) + a contract PDF (1 file). Use cost reports for hours/employees/supplies. Word doc for description/challenges/solutions. Contract for type/funded status. Do NOT create separate records for the Word doc or contract.

Expected Output: 1 record

17 Scenario E — Files With No Recognizable Project Data

User uploads files with no identifiable project: company policy doc, blank template, invoice for office supplies, HR file, generic spreadsheet. Do NOT hallucinate a record. Do NOT use company name or document title as project name.

Expected Output: 0 records

18 Scenario F — Files With Partial Project Data Only

User uploads files that reference a project but lack most fields — e.g., just a single invoice or just a title page. Create 1 partial record with populated fields shown and all others marked [Not found in uploaded documents]. Confidence: 40–60%. Never guess missing fields.

Expected Output: 1 partial record

19 Scenario G — One File Contains Multiple Projects

A single uploaded file covers multiple Job Numbers (e.g., a company-wide annual cost report). The 'Report Recap by Job' section at the end is a SUMMARY — do NOT create additional records from recap rows. Create one record per Job Number found in the data section only.

Expected Output: 1 per Job# found

20 Scenario H — Duplicate Files Uploaded (Same File Twice)

User accidentally uploads the same file twice (identical content, may have different filename). Detection: same Job Number + same totals + same date range = same file. Process it once. Hours are NOT doubled. Fields are NOT duplicated.

Expected Output: 1 record

21 Scenario I — Overlapping Date Range Files for Same Project

File 1: Job 314, Jan–Jun 2025 → 3,200 hrs. File 2: Job 314, Jan–Oct 2025 → 6,059 hrs. File 2 INCLUDES File 1's period. Use the broader range only. NEVER add 3,200 + 6,059 = 9,259 — that double-counts January through June.

Expected Output: 1 record, wider range

22 Scenario J — File Has Project Data But No Job Number

A Word document or older report describes a project by name only, no Job Number anywhere. Use the project name as the identifier. If other uploaded files have a Job Number matching this name → link them. Confidence: 60–70%. Flag: [No Job Number found — using project name as identifier].

Expected Output: 1 record (low conf.)

23 Scenario K — Scanned / Image-Only PDFs

User uploads a PDF that is a scanned paper document. No text layer. Text extraction returns nothing. Do NOT attempt to guess content. Flag as unreadable. Request re-upload of a text-selectable version.

Expected Output: 0 records, flag

24 Scenario L — Password-Protected Files

User uploads an encrypted or password-protected PDF or Word document. Cannot be opened without the password. Flag and request an unlocked version. Do not create any record.

Expected Output: 0 records, flag

25 Scenario M — Mixed Language / Foreign Language Documents

Files are partially or fully in a language other than English. Extract data regardless of language. Translate field values to English when populating the form. Note: 'Project name translated from [language]'. Confidence: 65–80%.

Expected Output: 1 record + note

PART 4 — DATA EXTRACTION RULES (7 Rules)

26 D-1 — Project Title: Use Content Identifier, Never Filename or Report Title

- Source of truth: Job: [number] in report header + job name in data section
- NOT the filename: '314_RD_JOB_PR_COST_DETAIL_103125.pdf'
- NOT the report title: 'Job Cost History Report From 01/01/25...'
- NOT any phase description: 'Phase 032030 Seal Paired Sheetpile'
- NOT the company name: 'Gordon N. Ball, Inc.'

27 D-2 — Total Man Hours: Labor Only, Include Overtime

- Count: PR (Payroll) transactions only — Regular + Overtime hours
- Do NOT count: EQ (equipment machine hours), AP (vendor invoices), GL/IC/JC/OH entries
- Verify: cross-check summed total against 'Total for job:' or 'Report Recap by Job' line
- Example: Regular 5,926.50 + OT 133.00 = Total 6,059.50 ✓ (not 5,926.50)

28 D-3 — Employees Completing Research: Unique Individuals Only

- Primary dedup key: Employee Code (AGUDAN, GILJES, GOMVIC)
- Same employee in 30 phases → count ONCE
- Do NOT count: AP vendor entries (F3 & Associates, Municon West Coast)
- Do NOT count: EQ equipment codes (CAT 308CR Excavator)

29 D-4 — Supplies Used: Materials and Equipment Costs

- Include: Cost Type M MATERIALS (dollar amounts only)
- Include: Cost Type E EQUIPMENT (rental/use costs)
- Exclude: PR Labor costs (that's man hours, not supplies)
- Exclude: S SUBCONTRACTOR costs (contracted services)

30 D-5 — Description Fields: Synthesize into Prose, Never List Phase Codes

- Collect all phase descriptions → group by activity type → write coherent paragraph
- Good: 'This project involved iterative design and hydraulic testing of flood infrastructure...'
- Bad: 'Phase 032030, Phase 038090, Phase 049030...'
- Applies to: Description, Solutions Considered, Technical Challenges

31 D-6 — R&D Qualification: Apply Four-Part Test Per Phase Category

- Qualifies: hydraulic testing, iterative design phases, novel repair under uncertainty
- Does not qualify: project management, cleanup/demob, landscaping, permitting
- Partial: project with mix of qualifying and non-qualifying phases → 'Partial — Needs Review'

32 D-7 — The Funded? Field: Signals from Contract Type

- Government agency as client → likely Funded
- Fixed-price, cost-plus contract → likely Funded
- Internal R&D; company bears all risk → likely Not Funded
- No contract info in files → mark Unknown, flag for manual entry

PART 5 — ABSOLUTE PROHIBITIONS (9 Rules)

These are hard rules. There are no exceptions and no overrides. They represent the nine most common failure modes.

33 NEVER create a record from a Phase description

A phase is a sub-task. 'Seal Paired Sheetpile Joint', 'Water Test Flood Door FW2', 'Project Superintendent', 'RESTORATION IMPROVEMENTS', 'FENCE ADDITION' are phase names — NEVER project names.

34 NEVER use the report title as the project name

'Job Cost History Report From 01/01/25 To 10/31/25' is the document type, not the project.

35 NEVER use the filename as the project name

'314_RD_JOB_PR_COST_DETAIL_103125.pdf' is a file naming convention, not the project.

36 NEVER use the company name as the project name

'Gordon N. Ball, Inc.' is the contractor/company, not the R&D; project.

37 NEVER create duplicate records for the same Job Number

If Job 314 appears in 10 files, the answer is always 1 record. Always.

38 NEVER create a record from the Report Recap section

The 'Report Recap by Job' or 'Total for Job:' section at the document end is a mathematical summary. It is not new data. It does not create new records.

39 NEVER count vendor or subcontractor names as employees

AP F3ASSO / F3 & Associates is a vendor payment. Only PR (Payroll) transaction codes are employees.

40 NEVER double-count hours from overlapping date ranges

If File A covers Jan–Jun and File B covers Jan–Oct for the same project, use File B's total only. File B already includes File A's data.

41 NEVER hallucinate data

If a field cannot be found in uploaded documents, mark it as [Not found in uploaded documents]. A blank field is always better than a wrong value on an R&D; tax credit claim.

PART 7 — CONFIDENCE SCORING (5 Tiers)

Score	Label	Conditions
90–100%	High	Identifier explicit; totals verified against recap; all fields from explicit data; consistent across files
75–89%	Good	Clearly identified; most fields populated; minor gaps (partial year, 1–2 inferred fields)
60–74%	Moderate	Identified from one source; significant fields inferred; date range partial
40–59%	Low	Identifier ambiguous; conflicting data; partially unreadable document
Below 40%	Insufficient	Cannot identify project — do NOT auto-create; flag for manual entry

PART 8 — EDGE CASE QUICK REFERENCE (22 Cases)

Situation	Expected Output	Records
10 files, all same project	1 record	1
10 files, 3 different Job Numbers	3 records — one per Job#	3
1 file containing 4 Job Numbers	4 records — one per Job#	4
File has no Job Number anywhere	Use project name as identifier; flag low confidence	1
Same file uploaded twice	Detect duplicate; process once; do not double hours	1
File 1 = Jan–Jun hours, File 2 = Jan–Oct same job	Use File 2 (broader range); never add both	1
Same phase name appearing 4 times	All belong to same project; roll into 1 record	1
Change order suffix in name	Fold into base job record; note scope expansion	1
Report Recap rows at end of document	Do NOT create records from recap; it's a summary	0 extra
Word doc with description but no numbers	Use for narrative; combine with cost report for numbers	1
Employee appears in 20 phases	Count them ONCE in employee total	—
Equipment hours alongside labor hours	Labor only for Man Hours; equipment costs for Supplies	—
Scanned PDF, no text layer	Flag as unreadable; request re-upload	0
Cost report is partial year only	Note date range; never invent full-year numbers	—
No R&D; project data in any file	0 records; explain what was found instead	0
Negative dollar amounts (credits)	Include in cost calculations; they net out naturally	—
Zero-hour phases (e.g. DOOR DESIGN ISSUES)	Include as context in description; exclude from hour totals	—

Subcontractor invoices with project-like names	AP type = vendor transaction; not separate projects	0 extra
'UnPosted?' label in PDF	UI artifact from accounting software; data is valid; ignore	—
Foreign language document	Extract and translate; note translation in flags	1
Similar phase name, different phase codes	Distinct phases; all belong to same project	1
Phase name truncated mid-word	Match by phase code number, not by truncated text	—

PART 9 — FINAL DECISION CHECKLIST (8 Questions)

Run this checklist before creating any record. All 8 questions must be answered.

42 Q1: Do I have a project identifier (Job#, Project#, Contract#, Product Name)?

→ If NO → use name as identifier; flag as low confidence

43 Q2: Is this identifier different from all identifiers I have already recorded?

→ If NO (duplicate) → merge into existing record; do NOT create new

44 Q3: Am I reading a Phase description, not a Job description?

→ Test: Is there a 4–6 digit numeric code before the name? → Phase → no record

45 Q4: Am I reading a Report Recap or Total row?

→ If YES → do NOT create a record; it is a summary of data already captured

46 Q5: Am I reading a company name, filename, or report title?

→ If YES → do NOT use these as the project name

47 Q6: Have I read ALL uploaded files before creating this record?

→ If NO → continue reading all files before creating any records

48 Q7: Will creating this record result in more records than actual projects?

→ If YES → stop; re-examine; you are likely splitting what should be merged

49 Q8: Do I have enough data to populate at least 3 fields with confidence?

→ If NO → create a partial record and flag for human review

PART 10 — UNIVERSAL PROJECT IDENTITY DETECTION (9 Rules)

These rules handle all industries and naming conventions — not just construction cost reports with 'Job:' headers. Projects can be named in dozens of ways.

77 Four-Tier Identifier Detection (Priority Order)

- Tier 1 — Explicit numeric/alphanumeric codes: Job:, Project #:, Contract:, Award #:, Grant #: (highest reliability)
- Tier 2 — Document title/heading after stripping document-type language
- Tier 3 — Name inferred from repeated proper noun (3+ occurrences) in body text
- Tier 4 — Nothing found → flag for manual entry; never hallucinate

78 Document-Type Phrases to Strip and Ignore

- Strip from titles: 'Technical Proposal:', 'Statement of Work:', 'Cost Report —', 'Payroll Report:'
- Strip: 'Job Cost History Report', 'Payroll Hours Analysis Report', 'Summary:', 'Phase Detail'
- After stripping: 'R&D; Cost Report — Next-Gen Turbine Blade' → 'Next-Gen Turbine Blade'

79 Cross-File Matching Without a Shared Number

- Step 1: Exact identifier match → definite same project
- Step 2: Normalized name match (lowercase, strip filler words) → same project
- Step 3: Semantic similarity only → FLAG both, never auto-merge
- Step 4: Same company + date range + scope → FLAG, request confirmation

80 Industry-Specific Document Patterns

- Construction: 'Job: 314' in report header → 'Job: 314 COYOTE FLOOD MANAGEMENT' in data row
- Engineering proposals: cover page title after stripping 'Technical Proposal:'
- Federal contracts: Contract No. + award title → both used
- Software/tech: product name or internal code on cover/header
- Biotech/pharma: protocol title or compound code (e.g., GNB-001)
- Manufacturing: product code or initiative name (e.g., TURB-NXT-2025)

81 Product Name IS the Project Name

- Common in: software (Nexus AI Engine), manufacturing (ThermaShield Coating), biotech (GNB-001)
- Product name appears consistently as proper noun across all files
- Multiple files about the same product = same project = 1 record

82 File Has Both a Code AND a Name

- Use code as deduplication key; name as display title; show both in output
- Example: 'R&D-2025-TURB-07;; Next-Gen Turbine Blade' → ID: R&D-2025-TURB-07;;, Name: Next-Gen Turbine Blade

83 Version Numbers and Iterations

- Different version numbers = different projects UNLESS same identifier or fully overlapping dates
- When uncertain: create separate records, flag as potential merge candidates for user to confirm

84

Ambiguous Codenames ('Project Alpha')

- Use codename as title placeholder
- Search all files for longer description of what the codename refers to
- If not found: flag description field as requiring manual entry

85

Semantic Similarity — Never Auto-Merge

- 'Autonomous Drone Bridge Inspection' + 'AI-Powered Bridge Inspection Platform' may or may not be same
- Create separate records; add flag: 'These projects have overlapping themes — please confirm if same project'

PART 11 — THE DEDUPLICATION ALGORITHM (4 Phases + Verification)

The step-by-step algorithm that mathematically guarantees the correct record count every time, regardless of how many files are uploaded or how complex the batch is.

86 Phase 1 — INVENTORY: Read Everything, Create Nothing

- For EVERY file extract exactly 5 keys: Identifier, Name, Company, Date Range, Report Type
- Create an inventory card per file. Zero records created in this phase.
- If file unreadable → mark UNREADABLE on card; continue with other files

87 Phase 2 — GROUPING: Cluster Files by Project Identity

- For each file: if it has an identifier that matches an existing group → add to that group
- If identifier is new → create ONE new group
- If no identifier but has name → match by normalized name; if no match → new group
- If no identifier and no name → mark UNIDENTIFIED; set aside
- After all files processed: group count = MAXIMUM possible records

88 Phase 2B Check 1 — Shared Identifier Prefix

- 'Job 314' + 'Job 314 - Fence Addition' → same identifier prefix → MERGE

89 Phase 2B Check 2 — Name is Subset of Another Group

- 'FloodGuard 3.0' + 'FloodGuard 3.0 Testing' + 'FloodGuard 3.0 Development' → core name match → MERGE all

90 Phase 2B Check 3 — Change Order Pattern

- Base name + scope suffix → fold into base group

91 Phase 2B Check 4 — Phase Description Accidentally Became a Group

- Test: does this group's name appear as a sub-item (phase) inside another group's documents?
- If YES → dissolve this group; reassign file to parent project group

92 Phase 2B Check 5 — Report Recap Row Accidentally Became a Group

- Text like '314 COYOTE FLOOD MANAGEMENT 6,059.50 \$572,728.26' in a footer → dissolve immediately

93 Phase 3 — EXTRACTION: Fill Fields Per Group

- Only after Phase 2B validation is complete
- Each group uses ONLY its own assigned files — never cross-group contamination
- Extract all 12 fields per group using Part 4 rules

94 Phase 4 + Verification

- Output exactly 1 record per group
- VERIFICATION: Records output MUST equal groups after Phase 2B
- If records > groups → false split exists; find and fix before delivering output
- If records < groups → missed a project; find the missing group

95 **Worked Example 1: 10 files, 10 different projects**

- Phase 1: 10 inventory cards, all distinct identifiers/names
- Phase 2: 10 groups formed
- Phase 2B: no false splits → 10 groups stand
- Output: 10 records ✓ Verification: $R(10) = G(10)$ ✓

96 **Worked Example 2: 10 files, same project**

- Phase 1: 10 inventory cards, all identifier '314'
- Phase 2: File 1 creates Group A; Files 2–10 all match '314' → added to Group A → 1 group
- Output: 1 record ✓ Verification: $R(1) = G(1)$ ✓

97 **Worked Example 3: 10 files, mix of 3 projects + 2 irrelevant files**

- Groups formed: 3. Unidentified: 2 (HR policy, blank template)
- Output: 3 records + notification about 2 irrelevant files ✓

98 **Worked Example 4: 1 file with 60+ phases (the dangerous case)**

- Phase 1: 1 card, identifier '314'
- Phase 2: 1 group
- Phase 2B Check 4: ALL 60+ phase descriptions dissolved
- Phase 2B Check 5: Report Recap rows dissolved
- Output: 1 record ✓ — despite 60+ potential false triggers

PART 12 — GUARDRAILS: HUMAN-IN-THE-LOOP PROTECTION (30 Rules)

CORE PRINCIPLE: Silence is better than a wrong answer. A blank field that a human fills in correctly causes zero harm. A wrong field submitted to an R&D; tax credit claim causes legal and financial harm. The agent fills fields it is CERTAIN about and flags everything else.

GLOBAL WRONG-FILE GUARDRAILS

99 Wrong-File Detection Test (3 Questions)

- Q1: Does this file contain a named project, job, product, or research initiative?
- Q2: Does this file contain hours, costs, employees, technical descriptions, or R&D; activities?
- Q3: Is the content specific to an identifiable entity (not generic policy)?
- All 3 NO → file contains no R&D; project data → exclude

100 All Files Fail the Test → 0 Records

- Output 0 records. Display explanation of what was found and what to upload instead.

101 Some Files Fail, Some Pass

- Process passing files. Flag excluded files with explanation.

102 All Files Ambiguous

- Output 0 records. Ask user to confirm before proceeding.

103 Files Belong to a Different Company's Project

- Flag: 'These files appear to reference [Company X]'s project. Please confirm this is correct.'

104 Files Have Financials But No Project Identifier

- Never invent a project from financial totals alone. Output 0 records.

PER-FIELD CONFIDENCE THRESHOLDS

Every field has a minimum confidence required to auto-fill. Below that threshold the field is left EMPTY with a flag — never filled with a guess.

Field	Auto-Fill Threshold	When Blocked — Default
Project Title	85%	Empty + flag
Contract Type	75%	'Unknown' + flag
Description	70%	Empty + flag
Qualification Status	80%	'Needs Review' + flag
Funded?	80%	'Unknown' + flag

Passes Four-Part Test?	80%	'Unknown' + flag
Total Man Hours	85%	Empty + flag
Employees Completing Research	80%	Empty + flag
Supplies Used	75%	Empty + flag
Solutions Considered	65%	Empty + flag
Technical Challenges	65%	Empty + flag
Contracts	85%	Empty (no flag — optional)

PER-FIELD GUARDRAIL RULES (105–116)

105 Project Title Guardrail

- Never auto-fills from: filename, report type, company name, phase description, recap row, footer
- 85%+ with explicit Job# + name → auto-fill ✓
- 70–84% name from doc title only → auto-fill + flag
- 50–69% multiple candidates → BLOCK; show all candidates; ask user to choose

106 Contract Type Guardrail

- 75%+ with explicit contract type → auto-fill ✓
- No contract document uploaded → auto-fill 'Unknown' + flag

107 Description Guardrail

- 70%+ with sufficient phases → auto-fill synthesized text + review flag
- Never includes phase codes, dollar amounts, or employee names
- Below 50% → BLOCK; leave empty

108 Qualification Status Guardrail

- Never auto-fills 'Qualified' when: only cost reports uploaded with no tech description
- Never auto-fills 'Qualified' when: all phases are administrative
- 80%+ with clear Four-Part Test evidence → auto-fill ✓

109 Funded? Guardrail

- No contract document in files → BLOCK; auto-fill 'Unknown' + flag

110 Four-Part Test Guardrail

- Linked to Qualification Status — if one blocked, both blocked

111 Total Man Hours Guardrail

- Never uses: EQ equipment hours, AP vendor hours, combined all-cost-type total
- Conflicting totals across files → BLOCK; show both values; ask user to clarify

112 Employees Guardrail

- Never counts: AP vendors, EQ equipment codes, same person twice across phases

113 Supplies Used Guardrail

- Partial data → auto-fill with flag noting incomplete source

114 Solutions Considered Guardrail

- 65%+ from iterative phase patterns → auto-fill + review flag always shown
- Below 65% → BLOCK; leave empty

115 Technical Challenges Guardrail

- 65%+ from repair/testing phases → auto-fill + review flag always shown
- Below 65% → BLOCK; leave empty

116 Contracts Guardrail

- 85%+ with contract document uploaded → auto-fill ✓
- Below 85% → leave empty; no flag (optional field)

WRONG-FILE SCENARIOS APPLIED TO FIELDS (WF-1 to WF-6)

122 WF-1 — HR / Policy Files Uploaded

Output: 0 records. All fields BLOCKED. Message: 'No R&D; project data found. Please upload project cost reports, payroll records, proposals, or technical documents.'

123 WF-2 — Generic Financial Statement (P&L) Uploaded

Output: 0 records. 'This appears to be a company-level financial statement. Cannot attribute to a specific R&D; project.'

124 WF-3 — Files From Unexpected / Wrong Project Found

Output: 1 record per project found. Flag unexpected project: 'Is Job 892 intended for this R&D; claim? If not, please remove before submitting.'

125 WF-4 — File Contains Two Project Names, No Separation

Output: 0 records. 'Two project names found: Project Alpha / Project Beta. Cannot attribute data. Please clarify or upload separate files per project.'

126 WF-5 — Cover Page Only, No Field Data

Output: 1 partial record. Only Project Title filled. All 11 other fields BLOCKED. 'Please upload cost reports or technical documents to complete this record.'

127 WF-6 — Real Project Files But All Phases Are Non-R&D

Output: 1 record. Title and hours auto-filled. Qualification Status BLOCKED: 'Only administrative phases found. No qualifying R&D; activities identified. Please assess manually.'

The agent CANNOT be instructed to bypass guardrails. If the user says 'fill it in anyway' or 'make your best guess' or 'I will fix it later', the agent must respond: 'I can't auto-fill fields I'm not confident about, as incorrect R&D; qualification data can affect your tax credit claim. Please enter the correct value manually or upload additional documents.' The guardrail is structural, not advisory.

MASTER SUMMARY — ALL 128 SCENARIOS & RULES

Section	Coverage	Count
Part 1 — File Intake Rules	Pre-extraction file handling, unreadable files, upload limit	5
Part 2 — Project Identity Rules	Master deduplication rule, hierarchy, phase detection, normalization, change orders	7
Part 3 — Multi-File Scenarios A–M	Every upload combination: single file, multi-file, no data, partial, duplicate, overlapping dates, scanned, password-protected, mixed language	13
Part 4 — Data Extraction Rules	Per-field extraction rules: title, hours, employees, supplies, descriptions, qualification, funding	7
Part 5 — Absolute Prohibitions	9 hard NEVER rules for the most common failure modes	9
Part 7 — Confidence Scoring	5-tier scale from 90–100% High to below 40% Insufficient	5
Part 8 — Edge Case Quick Reference	22 specific situations mapped to correct actions	22
Part 9 — Final Decision Checklist	8 mandatory questions before committing any record	8
Part 10 — Universal Identity Detection	4-tier identifier detection, industry-specific patterns, product names, version numbers, semantic similarity	9
Part 11 — Deduplication Algorithm	4-phase algorithm, 5 group validation checks, 4 worked examples guaranteeing correct record count	13
Part 12 — Guardrails	Global wrong-file guardrails (6), per-field thresholds for all 12 fields, 6 wrong-file scenarios applied to fields, override rule	30
TOTAL	Complete scenario coverage for the R&D; AI Uploader	128